

Investment Thesis on Defense ICT startups

Juho Kim

Seoul National University

juho.kim303@gmail.com

Investment Thesis: Market

[Target Market] Defense ICT

- **Defense**
 - Complete **systems** and critical **components** developed for national **defense and military purposes**; excluding MRO
- **Defense ICT**
 - **Defense systems and mission software that convert battlefield data into intelligence, decision support, and interoperable operations**

[Market Environment]

- **Favorable conditions are forming for operators, while broader investment opportunities are emerging for investors**
- **Public**
 - Governments are increasing defense spending and introducing policies to mobilize private capital into defense innovation
- **Private**
 - Rising defense demand is accelerating innovation, while emerging profitability potential is attracting more private investment

[Market Trend]

- **Changes in modern warfare are creating new capability demands and opportunities that non-legacy players can target**
- **Capability Demand**
 - 1) Affordable mass, 2) Advanced intelligence, 3) Interoperability
- **Opportunity**
 - 1) Dual-use adoption, 2) Interoperability fabric and application-module stacks

Investment Thesis: Scope & Focus

[Product Scope]

- (+) Systems and modular applications **supplied directly to end-users**
- (+) **Cost-effective** or **advanced-intelligence** systems compared with existing solutions
(OR) new **interoperability fabric / interoperable applications**
- (-) **Components supplied to primes** for large integrated systems; slow adoption cycle and prime-dependent growth

[Product Focus] Space ISR, decision/C2 AI, tactical data link, unmanned platforms, MUM-T modules, (anti-) jamming solutions, etc.

[Company Scope]

- VC-addressable companies that solve **capability gaps** by suggesting **new technological concept** or **benchmarking proven entry-and-scale strategies**
 - Capability gaps arise due to: **1) Localization needs, 2) Diverse mission requirements**
 - Companies proposing entirely unprecedented technology concepts carry higher uncertainty – selective approach needed
- Screening would rely hugely on **solution concept, defense-use cases, and demonstration references**
 - Detailed information is limited in public due to the nature of the defense industry

[Company Targeted]

- [IR Priority] 뉴타입인더스트리즈, 데이터메이커
- [Watchlist] 스페이스맵, 젠젠에이아이, Bone AI

Defense Industry: 1) Government-driven, 2) Mission-critical Systems Market where 3) Prime Contractors Orchestrate System Integration

- **Governments** define security capability requirements; **selected prime contractors** then design, integrate, deliver mission-ready systems
- **Commercial viability** requires either **winning government programs** as a “prime” or **supplying critical components** to primes
- Once embedded, suppliers can secure **long-term incumbency** through qualification and lifecycle lock-in

Value Chain

	Government	Prime Contractor	Suppliers	MRO
Planning	• Defense strategy setting • Requirements definition	• Early solution concept input • Technology feasibility input		
Program/Partner Selection	• RFP issuance • Partner selection	• Proposal commitment • Contract terms negotiation		
Dev., Integration & Testing	• Qualification / acceptance testing • Production approval	• System design / integration • Prototype development • Production ramp-up	• Components, subsystem supply	
Deployment & Upgrade	• Field deployment • Follow-on support contracting	• Final delivery • Initial operational support		• Maintenance, repair & overhaul • Life extension & upgrade execution
Mission Layers & Examples	Mission Layers	Integrated Systems	Critical Components & Enabling Technologies	
	ISR* / Sensor	Radar, satellite, drone, surveillance sys.	Sensors, signal collection, target detection, geospatial data, data processing	
	C4I* / Communications	C4I., communication networks, fire control sys., battlefield management sys.	Tactical communications, data links, sensor fusion, decision-support SW, mission planning tools	
	Precision Fires & Strike	Missiles, artillery, rockets, munitions	Guidance systems, seekers, targeting modules, warheads, fire-control SW	
	Physical Platforms	Fighter aircraft, naval vessels, UAVs	Body, protection, propulsion, navigation, autonomy, payload integration	
	EW*, Cyber & Spectrum Operations	EW sys., cyber defense sys., secure network sys., radar sys.	RF sensing, jamming, spectrum management, encryption, cyber threat detection, secure mesh networking	
	CBRN* / Special Threats	CBRN detection sys., base-protection	CBRN sensors, protective equipment, hazard analysis SW	

Source: 국방과학기술 표준분류체계, EY (Jul '25)

* ISR (Intelligence, Surveillance, and Reconnaissance) / C4I (Command, Control, Communications, Computers, and Intelligence) / EW (Electronic Warfare) / CBRN (Chemical, Biological, Radiological, and Nuclear)

Structural Change: Rising Defense Budgets are Shifting toward New Warfare Capabilities

- **Rising defense budgets** are not only expanding legacy procurement, but also creating new demand for modern warfare capabilities
- Governments are **strengthening domestic** defense ecosystems and **widening entry channels** for non-traditional defense players
- Emerging opportunities are for startup that target **new warfare capability gaps** and follow effective entry and scaling strategies

Structural Shift

Defense Budget Expansion

Structural rearmament under prolonged geopolitical instability

[Defense Budget Expansion] Long-term growth in defense spending

- Military expenditure; Rising in both absolute spend and GDP share commitment

[Domestic Production] Policy-driven build-up of local defense industrial capacity

[Startup Support] Policy-backed defense-tech startup ecosystem

- Expanding in support from both public and VC-backed private funding

Demand Shift

Shift toward asymmetric, technology-intensive, multi-domain warfare capabilities

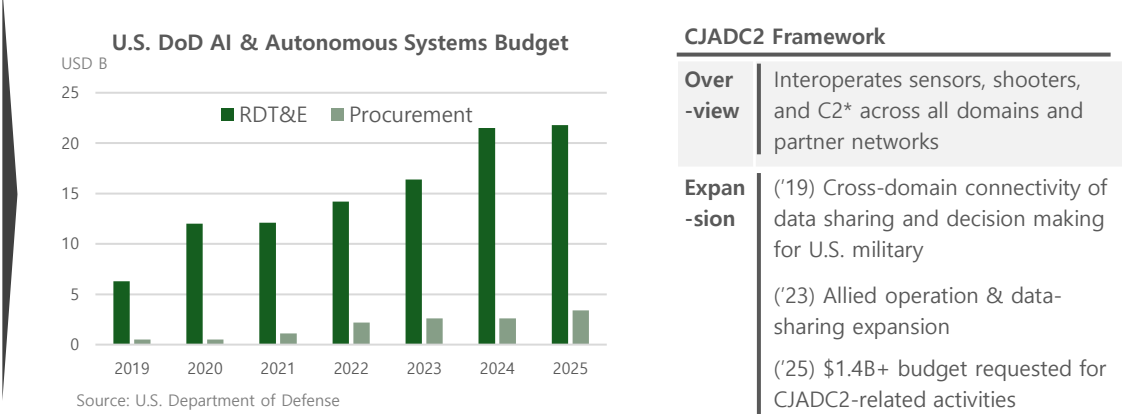
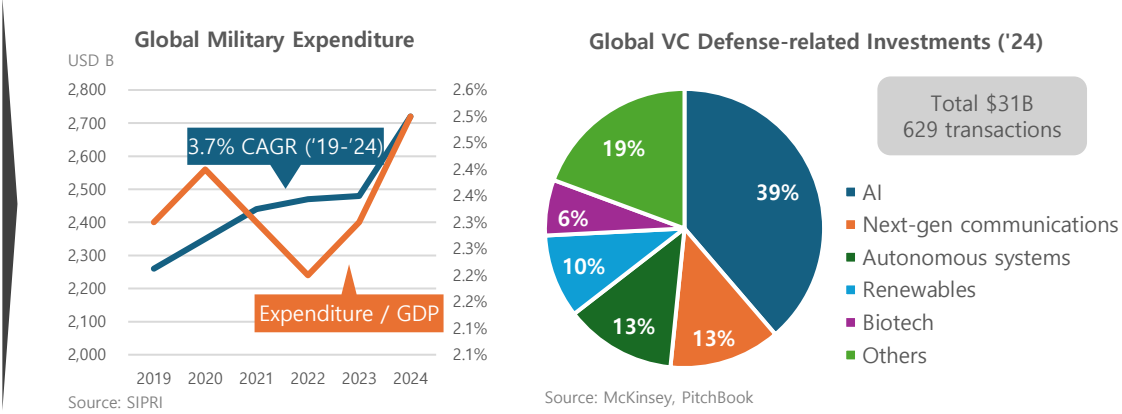
[Affordable Mass] Low-cost, scalable, quickly replenishable systems

[Advanced Intelligence] AI, autonomy, cyber/EW, and simulation adoption

- AI & Autonomous systems; Expanding US DoD*'s exclusive budget

[Interoperability] Modular architecture and system connectivity

- CJADC2* framework; Expanding in both budget scale and connectivity scope

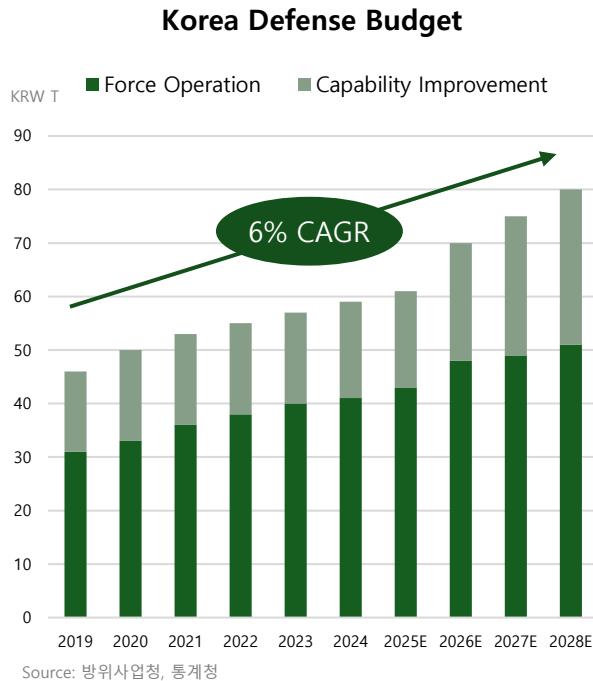


Source: SIPRI, EY
* DoD (Department of Defense) / CJADC2 (Combined Joint All-Domain Command and Control) / C2 (Command and Control)

Defense Budget Expansion: Korea is Funding Startup Ecosystem Growth and Private Capital Formation

- Korea is expanding defense spending, with **capability improvement budget** expected to grow faster than force operation spending
- Policy support is moving beyond traditional procurement, combining **direct startup ecosystem support** with **indirect financing**
- Environment becomes more favorable for both operator and investor, with **lower entry barriers** and **stronger private-capital incentives**

Defense Budget Plan



Defense-tech Innovation Plans

Defense Startup Promotion Plan / 방산 스타트업 육성방안

Over-view	[Purpose] Dual-use tech adoption and startup entry into defense ecosystem [Target / Lead] Dual-use startups / MSS*, DAPA* [Goal] 100 defense startups, 30 unicorn firms by 2030; particularly in AI, drones, robotics and other advanced dual-use fields
Pro-cess	Startup discovery > Military/prime feedback > Field testing > Acquisition linkage > Growth support
Key support	[Market Entry Support] PoC, field demonstration, operational feedback, acquisition linkage [Growth package] Policy financing, investment IR, regulatory/IP advisory, oversea exhibition [Infrastructure] Access to defense data, test facilities

Ecosystem Vitalization
Lower entry barrier
+
Startup-inclusive defense-tech innovation

Defense Technology Innovation Fund / 방산기술혁신펀드

Over-view	[Purpose] Defense-tech investment and private capital mobilization [Target / Lead] GPs investing in defense and dual-use tech firms / DAPA, KGrowth* [Goal] KRW 310B fund; crowd-in private capital for defense strategic fields such as space, AI, drones
Pro-cess	GP selection > Private capital matching > Investment into defense / dual-use technology firms
Key support	[Anchor LP capital] Policy capital committed to external GPs [Defense-tech mandate] Dedicated allocation into defense and dual-use strategic fields

Private Investment Stimulation
Anchor capital supply
+
Defense-tech investment mandate

Demand Shift: New Warfare Creates Demand for Affordable Mass, Advanced Intelligence, and Interoperable Modules

- New warfare is reshaping capability demand – **affordable mass**, **advanced intelligence**, and **interoperable modules**
- **Intelligent technologies** are increasingly adopted into military systems, deriving much higher demand for **dual-use** technology
- Startup opportunities are emerging over reshaping **defense-tech stacks**, especially related to **interoperability**

New Capability Needs

Capability	Requirement
Affordable Mass	• Low-cost, scalable, and rapidly replenishable systems • For sustaining attrition-heavy operations
Advanced Intelligence	• Real-time, AI-enabled intelligence loop and infrastructure systems • For converting distributed battlefield data into analysis, decision support and autonomous action
Interoperable Modules	• Modular, platform-agnostic, data-connected systems • For cross-platform, cross-domain, and allied operations

New Opportunity

Public Agencies Supporting Dual-Use Tech		
U.S.	DIU*; In-Q-Tel; CIA-led VC	Competitions with private companies holding dual-use technologies
U.K.	DASA*	Public funding for firms' proposal-based technologies applications
Israel	Innofence; defense R&D	Co-develop technologies with private data

Future Defense-tech Stack	Layer 5 Application Layer	AI and analytics solutions; currently siloed on disconnected tech stacks
	Layer 4 Interoperability Fabric	Common operating system enabling platform-agnostic applications; currently limited by lack of reference architecture
	Layer 3 Transport Mesh	Interconnected network transmitting data across systems
	Layer 2 Compute Foundation	Onboard compute hardware
	Layer 1 Physical Edge	Platform hardware across domains; shifting toward attritable, modular, mass-producible form factors

Dual-use Adoption	• Non-defense tech firms' entry into defense industry • Driven by defense-relevant demands for intelligent technologies • Active discovery and adoption of commercial technologies by government agencies
Interoperability Stack	• Modularization of integrated system • Driven by multi-domain operations and higher technology requirements • Rising need for the missing interoperability fabric and high-spec modules that connect platforms, data, and applications

Source: 매경이코노미, 국회 미래연구원, McKinsey
* DIU (Defense Innovation Unit), DASA (Defense and Security Accelerator)

Source: McKinsey

Startup Entry Points: Capability Gaps Create Opportunities for Startups, while Capability Demand Differs by Mission Layer

- New warfare demand varies by mission layer, from ISR analytics to autonomous platforms and cyber/EW enablement
- **Startup entry points** are concentrated where **affordability**, **advanced intelligence**, or **interoperability gaps** are strongest
- Screening should focus on **concrete product cases**, not broad defense categories

Missions & Needs

Mission Layer	Requirement & Rationale	Product / Service
ISR / Sensor	[Advanced Intelligence] Expansion of intelligence assets across space, and distributed sensors [Interoperable Modules] Multi-sensor fusion, and integrated processing and analysis	Space ISR, Specialized sensor modules AI ISR analytics, Sensor fusion modules
C4I / Communications	[Advanced Intelligence] AI-enabled large-scale data intelligence aggregation for decision support, battlefield management [Interoperable Modules] Interoperability fabric and transport mesh enabling cross-domain data-sharing and sensor-to-shooter interoperation	Decision AI / C2 AI Defense data platform, Tactical data link / communications
Fires / Precision Strike	[Affordable Mass] Attritable, low-cost, and scalable strike/interception assets beyond legacy large-scale firepower systems	Loitering drones / munitions, Counter-drone interceptors, Low-cost precision effectors
Platforms	[Affordable Mass] Attritable and mass-producible platforms beyond legacy large-platforms [Advanced Intelligence] UAV, autonomous navigation, robotics, and manned-unmanned teaming	UAV platforms, Robotics, Autonomous navigation, MUM-T* support modules
EW / Cyber	[Advanced Intelligence] AI-enabled threat detection, cyber intelligence, resilient communication systems, and automated response	Cyber threat intelligence, Jamming / anti-jamming solutions
CBRN / Special Threats	[Limited] Specialized demand, limited operating environments, high certification burden, and strong government-procurement dependence	n.a

U.S. Cases: Growth Paths Vary from **core AI Stack Lock-in** to advanced platform-integrated SW capabilities

- Successful cases often start with **innovative software**, then expand into **core SW layers** or own **advanced or cost-effective platforms**
- **Proven potential dual-use players** can enter defense market, utilizing capabilities developed and scaled outside the defense market
- **Proven playbooks** can be replicated in **adjacent capability gaps** where legacy systems are underperforming or missing

U.S. Defense-related Startups

	Palantir	SpaceX	Anduril	Shield AI	Saronic
Overview	[Founded] 2003. [Value.] \$367.9B (Public)	[Founded] 2002 [Value.] \$800B (Secondary '25)	[Founded] 2017 [Value.] \$30.5B (Series G '25)	[Founded] 2015 [Value.] \$12.7B (Series G '26)	[Founded] 2022 [Value.] \$9.25B (Series D '26)
Products	[Gotham] - Defense data platform, Decision AI - Mission data ontology for intelligence/defense operation [Maven Smart System] - AI ISR analytics, Decision AI - Multi-source ISR aggregation for decision support	[Starlink] - Data link, communication - Low-cost, resilient connectivity with scaled LEO constellation [Starshield] - Space ISR, defense satellites - New ISR capabilities; scaled LEO constellations utilization	[Lattice] - C2 AI, Interoperability fabric - Sensor-to-effector connectivity for battlefield management [Roadrunner] - Low-cost precision effectors - Reusability, attritability [Ghost / Altius] - ISR / loitering drones - Interoperability, autonomy	[Hivemind] - Autonomous navigation, MUM-T support modules - Platform-agnostic AI pilot; resilient capabilities [V-BAT] - Drone platform, AI ISR analytics - Interoperability; with Hivemind, other modules	[Corsair/Marauder] - Naval ISR, ASV platforms, MUM-support T modules - Interoperability, new ISR capability, scalable production
Entry & Scale	[SW module > Core SW] - Entered through analytics and data-integration layer - Expanded into broader C2 / AI workflows with superior technology concept	[Dual-use adoption] - Scaled infra. for commercial use with private funding - Adopted for defense uses to fill capability gaps	[SW-HW integrated system] - Proposed solutions to urgent pain points with innovative SW - Connected to cost-effective hardware systems - Bypassed slow pilot and adoption cycles	[SW module > HW platform] - Entered through innovative SW - Expansion into own HW platforms and other SW layers	[Benchmark; Core SW > HW] - Benchmarked proven autonomy platform playbooks to naval domain - Provided innovative SW with connected HW platforms

Non-U.S. Cases: Commercial Products Adapted for Military Use Created Defense Entry, Followed by **Stack Penetration** and **Scale Expansion**

- **Commercial or dual-use products** often served as the entry point, before being **adapted** to meet **military capability** requirements
- Winners **scaled** beyond entry products into **adjacent capability gaps**, especially **advanced-intelligence stacks** such as AI-powered C4I, simulations, autonomous/unmanned control platforms

Non-U.S. Defense-related Startups

	Helsing (Germany)	Hadean (UK)	ICEYE (Finland)	Quantum Systems (Germany)	DJI (China)
Overview	[Founded] 2021 [Value.] ~€12B (Series D '25)	[Founded] 2015 [Value.] n.a	[Founded] 2014 [Value.] €2.4B (Series E '25)	[Founded] 2015 [Value.] €3B+ (Series C '25)	[Founded] 2006 [Value.] n.a
Products	[Altra] - Decision AI, data platform - AI-enabled battlefield data fusion for decision support [HX-2] - Loitering munitions, autonomous platforms - Cost-effective; advanced autonomous system for contested ops. Environment	[PopulAI / DominAI / OptimAI] - Decision AI, C2 AI, AI ISR analytics for simulation - Data synthesis; advanced C4I intelligence in simulation [SCEPTRE] - AI-powered digital wargame platform - New capability for large-scale training and mission rehearsal	[Microsatellites Constellation] - Space ISR, specialized sensor modules - SW-defined modularity, scalability [Intelligence Services] - AI ISR analytics, geospatial intelligence - Persistent, large-scale intelligence and analytics	[Vector / Scorpion] - ISR drones, UAV platforms - EW/cyber-resilient capabilities [Jaeger] - AI-powered autonomy system - Edge processing [MOSAIC UXS] - Unmanned mission control SW - Multi-domain MUM-T capabilities	[Mavic / Mini / Air / Matrice] - Low-cost UAV platform, ISR drones - Attritability, modularity [DJI Enterprise Payloads] - Functional modules such as AI ISR analytics, fire/strike - Modularity; with distributed and remote control
Entry & Scale	[Core SW > HW platform] - Entered through core C4I AI SW, aligning with regional defense demand - Expanded into autonomous HW systems connected with SW	[Dual-use adoption] - Scaled as spatial-computing entertainment product - Adopted by MoD providing computing and simulation infra. through partnership within ecosystem	[Dual-use adoption] - Entered through commercial satellite-based info. services - Staged scale-up to military use and constellation-scale services	[Dual-use adoption] - Integrated military capabilities and autonomous systems on commercial drones, addressing urgent EU ISR demand - Expanded toward broader MUM-T mission control system	[Dual-use adoption] - Scaled as commercial B2C/B2B drone business - Repurposed by military end-user; cost-efficiency - Added military payloads on identical HW platform

Investment Target: VC-Scale Opportunities are Concentrated in ISR / C4I Software Players Already Building Defense References

- **ISR / C4I software** remains the most **VC-addressable segment**, with opportunities across various mission layers and capabilities
- Some specialized software firms have already been captured by primes, through **strategic investment** or acquisition
- **Hardware platforms** are **less attractive** as new VC targets due to accumulated funding, IPO tracks, or strategic-investor ownership

VC opportunities in Korea market

Company	Product Focus	Differentiated Capability	Entry & Scale
씨드로닉스	Maritime autonomy platform USV AI navigation monitoring system (NAVISS)	[Advanced Intelligence] AI-powered sensor fusion for autonomous navigation and monitoring in maritime environment [Data] Large, port-to-port AI ops. Data of its commercial use-cases [Partnership] Dual-use partnership with Korean MoD	- Maritime autonomy entry; commercial navigation AI adapted to naval USV operation - Scale potential in expansion toward core C4I AI, port-fleet connectivity stacks
데이터메이커	Edge AI ISR, BDA* system (Synapse-Defense)	[Advanced Intelligence] On-device AI ISR, BDA capability [Interoperability] Integration with MLOps, data pipelines, platforms [Reference] Partnership with KIDA*, KISTI*, ALC*	- Entry through edge AI ISR gap; faster battlefield interpretation beyond fixed pre-trained object classes - Scale potential in integration with broader platforms / systems
스페이스맵	Space ops. support / simulation (SpaceTube)	[Advanced Intelligence] Digital twin for orbital-object and space-event analysis [Data/algorithm] Specialized for space ops. and monitoring [Reference] U.S. Space Force service-delivery reference	- Slingshot Aerospace-like solution concept - Entered through advanced space-domain operations support
젠젠에이아이	Synthetic data for defense AI training	[Advanced Intelligence] Synthetic data for rare, unsafe scenarios [Knowledge] Knowledge for high-quality synthetic data generation [Project] KAI / ADD* project on automatic target-identification task	- Entered as a simulation-data layer; military training-data scarcity bottleneck - Scale potential across broader mission areas and tasks
뉴타입 인더스트리즈	C2 AI; target processing and tactical fire-command (Barbara)	[Advanced Intelligence] Target identification-to-fire decision automation for artillery; connecting voice-based field report [Domain Expertise] 12-year artillery officer founder [Project] Field PoC with U.S. Army; Selected partner for fast track	- Palantir-like concept; Entered through core C2 AI entry; tactical fire-command workflow digitization - Scale potential from artillery target processing to broader sensor-to-shooter C2 modules
Bone AI	Full-stack unmanned AI systems; C2 AI with robot HW	[Advanced Intelligence] Physical AI across unmanned platform areas [Full-stack] Robotics HW with C2 AI, chips and manufacturing [Reference] Revenue-generating reference cases	- Anduril-like entry; full-stack autonomous systems for government and defense-adjacent missions

Source: 씨드로닉스, 데이터메이커, 스페이스맵, 젠젠에이아이, 뉴타입인더스트리즈, Bone AI, Platum

* BDA (Battle Damage Assessment), ADD (국방과학연구소), KIDA (한국국방연구원), KISTI (한국과학기술정보연구원), ALC (육군 군수창)

Investment Target: Defense-Tech Startups Remain **Early in Validation**, but **Capital Formation is Accelerating**

- Meaningful defense-segment **profitability** is **not yet visible**, making **PoC** and field testing **critical**
- Two 2025-founded C2 AI startups are **moving quickly**, raising seed **funding** and entering **demonstration-track** projects within a year
- Investment thesis should be updated through **IR**, verifying **technological foundation**, **use-cases**, and **partnership status**

VC opportunities in Korea market

Company	Overview	Investment History	Competition
씨드로닉스	[Founded/Founder] 2015/ PhD; autonomous-driving [Rev./Profit] ₩1.25B/-₩2.94B	Total ₩25B+ [Series A] 2021; ₩4.5B; SBVA, etc. [Series B] 2025; ₩15B; LB인베스트먼트, etc.	[Competitors] (Startup) Orca AI; (Legacy) Avikus – HD Hyundai [Edge] Commercial operating-data base, solution transferability [Weak] Matured competition, legacy ship-manufacturers' proprietary development
데이터메이커	[Founded/Founder] 2018/ ex-founder [Rev./Profit] ₩5.66B/-₩800M	Total ₩0.7B [Seed] 2020; ₩0.7B; 신용보증기금	[Competitors] (Foreign) Scale AI, RAIC Labs; (Domestic) 편진, 마크애니 [Edge] On-device zero-shot ISR analysis capability [Weak] Less proof on defense-use performance
스페이스맵	[Founded/Founder] 2021/ professor; space-traffic- management research [Rev./Profit] ₩180M/n.a.	Total \$1M [Seed] n.a.; 더인벤션랩, 인라이트벤처스	[Competitor] (Entrant) Slingshot Aerospace; (Incumbent) Lockheed Martin, L3Harris [Edge] Domestic sovereignty demand [Weak] Locality; limited adjacent coverage
젠젠에이아이	[Founded/Founder] 2022/ PhD; image-restoration [Rev./Profit] ₩620M/-₩1.06M	Total ₩17.3B [Pre-A] 2023; ₩4.5B; 스마일게이트인베., etc. [Series A] 2025; ₩12B; (SI) KAI, etc.	[Competitors] AI verse, Sky Engine AI [Edge] Synthetic data utility for rare, dangerous scenarios; KAI/ADD use reference [Weak] Less proof on domain performance; limited market expansion due to SI
뉴타입 인더스트리즈	[Founded/Founder] 2025/ ex-artillery officer at combined-force	Total n.a. [Seed] 2025; n.a.; 블루포인트파트너스, etc.	[Competitor] (Entrant) Palantir, Anduril; (Incumbent) C4I, fire-control primes [Edge] Field-level workflow with lower friction [Weak] Early product maturity
Bone AI	[Founded/Founder] 2025/ MarqVision co-founder [Rev./Profit] ~\$3M/n.a.	Total \$12M [Seed] 2025; \$12M; Third Prime, (SI) Kolon	[Competitors] (Full-stack) Anduril, Shield AI; (Platforms-focused) Ghost Robotics, Saronic [Edge] Full-stack capability, SI's vertical adjacency in value chain, early B2G revenue signal [Weak] Unclear performance, broad scope execution risk

Investment Thesis: Target

[IR Priority]

- 뉴타입인더스트리즈
 - **Attractive** advanced-intelligence **solution** addressing friction between field and control system on tactical-fires workflow
 - Likely to **require new funding** to **accelerate demonstration** projects and **early-stage expansion**
- 데이터메이커
 - **Growing demand** for on-device intelligence processing and decision support as autonomous platforms and weapon systems emerging
 - Potential **need for growth capital** while **adapting** systems for military deployment

[Watchlist]

- 스페이스맵
 - Needs to prove **expansion potential** beyond domestic users into overseas or **allied markets**
- 젠젠에이아이
 - Needs to prove the **effectiveness** of synthetic data in defense and **military-use domains**
- Bone AI
 - **Attractive concept** of integrated UAV, UGV, and USV C2 system, with early B2G revenue signal
 - UAV-focused yet; requires further evidence for **feasibility** of the unproven **concept**

[No More Review]

- 씨드로닉스
 - Late-stage profile, but still small revenue scale and exposed to mature competitors